§8 变分法

§8.1 量子力学的变分原理

能量 $E[\psi] = \dfrac{\langle\psi|\hat{H}|\psi\rangle}{\langle\psi|\psi\rangle}$，基态能量 E_0

$$E[\psi] \geqslant E_0$$

$$E[\psi] = \frac{\sum_n E_n |\langle n|\psi\rangle|^2}{\sum_n |\langle n|\psi\rangle|^2} \geqslant \frac{\sum_n E_0 |\langle n|\psi\rangle|}{\sum_n |\langle n|\psi\rangle|^2}$$

Ritz方法：$|\psi\rangle = \sum_i \alpha_i |i\rangle$

$$\frac{\partial E}{\partial \alpha_i} = 0$$

例：用变分法求解一维谐振子的基态.

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2} m\omega^2 \hat{x}^2$$

$$\langle x|\psi\rangle = A e^{-bx^2}$$

$$E(b) = \frac{\langle\psi|\hat{H}|\psi\rangle}{\langle\psi|\psi\rangle} = \frac{\hbar^2}{2m} b + \frac{m\omega^2}{8b}$$

$$\frac{\partial E(b)}{\partial b} = 0 \rightarrow b = \frac{m\omega}{2\hbar}, \quad E_0 \approx \frac{1}{2}\hbar\omega$$

$$\langle x|\psi\rangle = \left(\frac{2b}{\pi}\right)^{1/4} e^{-\frac{m\omega}{2\hbar}x^2}$$

§ 8.2 氦原子

$$\hat{H} = -\frac{\hbar^2}{2m}(\nabla_1^2 + \nabla_2^2) - \frac{e^2}{4\pi\varepsilon_0}\left(\frac{Z}{r_1} + \frac{Z}{r_2} - \frac{1}{|r_1 - r_2|}\right)$$

忽略电子的相互作用：

基态：$\phi_0(\vec{r}_1, \vec{r}_2) = \psi_{100}(\vec{r}_1)\,\psi_{100}(\vec{r}_2) = \frac{Z^3}{(\pi a)^3} e^{-\frac{Z(r_1+r_2)}{a}}$

代回计算能量，

$$\langle \hat{H} \rangle = 2Z^2 E_1 + \frac{e^2}{4\pi\varepsilon_0}\int \phi_0^2(\vec{r}_1, \vec{r}_2)\frac{1}{|\vec{r}_1 - \vec{r}_2|}\, d^3r_1\, d^3r_2$$

$$\frac{1}{|\vec{r}_1 - \vec{r}_2|} = \frac{1}{\sqrt{\vec{r}_1^2 + \vec{r}_2^2 - 2\vec{r}_1\vec{r}_2\cos\gamma}} = \begin{cases} \sum_l \frac{r_1^l}{r_2^{l+1}} P_l(\cos\gamma) & 0 \le r_1 \le r_2 \\ \sum_l \frac{r_2^l}{r_1^{l+1}} P_l(\cos\gamma) & r_1 > r_2 \end{cases}$$

$$\int_{-1}^{1} P_m(x) P_n(x)\, dx = \frac{1}{2n+1}\delta_{mn}, \quad P_0(x) = 1$$

$$\int \phi_0^2(\vec{r}_1, \vec{r}_2)\frac{1}{|\vec{r}_1 - \vec{r}_2|}\, d^3r_1\, d^3r_2$$

$$= \int \psi_{100}(r_2)\left[\frac{1}{r_2}\int_0^{r_2}\psi_{100}^2(r_1)\, d^3r_1 + \int_{r_2}^{\infty}\frac{\psi_{100}^2(r_1)}{r_1}\, d^3r_1\right] d^3r_2$$ 只有 $l=0$ 有贡献

$$= \frac{5Z}{8a}$$

考虑屏蔽效应，假设1个电子屏蔽后的核电荷为 $Z-\eta$

$$\hat{H} = -\frac{\hbar^2}{2m}(\nabla_1^2 + \nabla_2^2) - \frac{e^2}{4\pi\varepsilon_0}\left(\frac{Z-\eta}{r_1} + \frac{Z-\eta}{r_2} + \frac{\eta}{r_1} + \frac{\eta}{r_2} - \frac{1}{|r_1 - r_2|}\right)$$

变分波函数：$\phi_0(r_1, r_2) = \frac{(Z-\eta)^3}{\pi a^3} e^{-\frac{(Z-\eta)(r_1+r_2)}{a}}$

$$\langle \hat{H} \rangle = 2(Z-\eta)^2 E_1 - \frac{5}{4}(Z-\eta)E_1 - \frac{\eta e^2}{4\pi\varepsilon_0}\left(\left\langle\frac{1}{r_1}\right\rangle + \left\langle\frac{1}{r_2}\right\rangle\right)$$

$$\frac{\eta e^2}{4\pi\varepsilon_0}\left\langle\frac{1}{r_1}\right\rangle = \frac{\eta e^2}{4\pi\varepsilon_0}\left(\left\langle\frac{1}{r_2}\right\rangle\right) = -2\eta(Z-\eta)E_1$$

$$\langle \hat{H} \rangle = 2(Z-\eta)^2 E_1 - \frac{5}{4}(Z-\eta)E_1 + 4\eta(Z-\eta)E_1$$

以 η 为参数变分：$\frac{\partial\langle \hat{H} \rangle}{\partial\eta} = 0$ 解得 $\eta = \frac{5}{16}$

$$\langle \hat{H} \rangle = \frac{729}{128}E_1, \quad Z = \frac{27}{16}$$